

A Technical Introduction to Mathematical optimization

Section 1

This denotes the minimum value of the objective function $x^2 + 1$, when choosing x from the set of real numbers $\mathbb{R}$. The minimum value in this case is 1, occurring at $x = 0$. Similarly, the notation

Section 2

Existence The extreme value theorem of Karl Weierstrass states that a continuous real-valued function on a compact set attains its maximum and minimum value. More generally, a lower semi-continuous function on a compact set attains its minimum; an upper semi-continuous function on a compact set attains its maximum point or view.

Section 3

The iterative methods used to solve problems of nonlinear programming differ according to whether they evaluate Hessians, gradients, or only function values. While evaluating Hessians (H) and gradients (G) improves the rate of convergence, for functions for which these quantities exist and vary sufficiently smoothly, such evaluations increase the computational complexity (or computational cost) of each iteration. In some cases, the computational complexity may be excessively high. One major criterion for optimizers is just the number of required function evaluations as this often is already a large computational effort, usually much more effort than within the optimizer itself, which mainly has to operate over the N variables. The derivatives provide detailed information for such optimizers, but are even harder to calculate, e.g. approximating the gradient takes at least $N+1$ function evaluations. For approximations of the 2nd derivatives (collected in the Hessian matrix), the number of function evaluations is in the order of N^2 . Newton's method requires the 2nd-order derivatives, so for each iteration, the number of function calls is in the order of N^2 , but for a simpler pure gradient optimizer it is only N . However, gradient optimizers need usually more iterations than Newton's algorithm. Which one is best with respect to the number of function calls depends on the problem itself.

Section 4

Mechanics Problems in rigid body dynamics (in particular articulated rigid body dynamics) often require mathematical programming techniques, since you can view rigid body dynamics as attempting to solve an ordinary differential equation on a constraint manifold; the constraints are various nonlinear geometric constraints such as "these two points must always coincide", "this surface must not penetrate any other", or "this point must always lie somewhere on this curve". Also, the problem of computing

contact forces can be done by solving a linear complementarity problem, which can also be viewed as a QP (quadratic programming) problem. Many design problems can also be expressed as optimization programs. This application is called design optimization. One subset is the engineering optimization, and another recent and growing subset of this field is multidisciplinary design optimization, which, while useful in many problems, has in particular been applied to aerospace engineering problems. This approach may be applied in cosmology and astrophysics.

Section 5

Operations research Another field that uses optimization techniques extensively is operations research. Operations research also uses stochastic modeling and simulation to support improved decision-making. Increasingly, operations research uses stochastic programming to model dynamic decisions that adapt to events; such problems can be solved with large-scale optimization and stochastic optimization methods.

Section 6

Control engineering Mathematical optimization is used in much modern controller design. High-level controllers such as model predictive control (MPC) or real-time optimization (RTO) employ mathematical optimization. These algorithms run online and repeatedly determine values for decision variables, such as choke openings in a process plant, by iteratively solving a mathematical optimization problem including constraints and a model of the system to be controlled.

Section 7

For unconstrained problems with twice-differentiable functions, some critical points can be found by finding the points where the gradient of the objective function is zero (that is, the stationary points). More generally, a zero subgradient certifies that a local minimum has been found for minimization problems with convex functions and other locally Lipschitz functions, which meet in loss function minimization of the neural network. The positive-negative momentum estimation lets to avoid the local minimum and converges at the objective function global minimum. Further, critical points can be classified using the definiteness of the Hessian matrix: If the Hessian is positive definite at a critical point, then the point is a local minimum; if the Hessian matrix is negative definite, then the point is a local maximum; finally, if indefinite, then the point is some kind of saddle point. Constrained problems can often be transformed into unconstrained problems with the help of Lagrange multipliers. Lagrangian relaxation can also provide approximate solutions to difficult constrained problems. When the objective function is a convex function, then any local minimum will also be a global minimum.

There exist efficient numerical techniques for minimizing convex functions, such as interior-point methods.

Section 8

Besides (finitely terminating) algorithms and (convergent) iterative methods, there are heuristics. A heuristic is any algorithm which is not guaranteed (mathematically) to find the solution, but which is nevertheless useful in certain practical situations. List of some well-known heuristics:

Section 9

Mathematical optimization (alternatively spelled optimisation) or mathematical programming is the selection of a best element, with regard to some criteria, from some set of available alternatives. It is generally divided into two subfields: discrete optimization and continuous optimization. Optimization problems arise in all quantitative disciplines from computer science and engineering to operations research and economics, and the development of solution methods has been of interest in mathematics for centuries. In the more general approach, an optimization problem consists of maximizing or minimizing a real function by systematically choosing input values from within an allowed set and computing the value of the function. The generalization of optimization theory and techniques to other formulations constitutes a large area of applied mathematics.

Section 10

$\arg \max_{x \in [-5, 5], y \in \mathbb{R}} x \cos y, \{\underset{x \in [-5, 5], y \in \mathbb{R}}{\operatorname{arg\,max}}\} x \cos y,$

Section 11

Feasibility problem The satisfiability problem, also called the feasibility problem, is just the problem of finding any feasible solution at all without regard to objective value. This can be regarded as the special case of mathematical optimization where the objective value is the same for every solution, and thus any solution is optimal. Many optimization algorithms need to start from a feasible point. One way to obtain such a point is to relax the feasibility conditions using a slack variable; with enough slack, any starting point is feasible. Then, minimize that slack variable until the slack is null or negative.